Lecture 27.01: Problem Statement & Credit Risk Fundamentals

📌 Topic Overview

In this lecture, Shraddha ma'am introduces the **CreditWise Loan Approval System** minor project for **SecureTrust Bank**, detailing the business challenges of credit underwriting and why financial classification requires prioritizing **Precision** over raw Accuracy.

🏦 Business Context: SecureTrust Bank

A retail financial institution receives hundreds of daily loan applications for personal and home loans. Historically, human credit officers evaluated applications manually. This created two critical vulnerabilities:

1. **High Turnaround Time (TAT)**: 5 to 10 days per decision.
2. **Asymmetric Error Costs**:
    - **Type I Error (False Positive)**: Approving an applicant who defaults → **Severe principal loss (₹1,00,000+)**.
    - **Type II Error (False Negative)**: Rejecting a creditworthy applicant → **Mild lost interest opportunity (₹10,000)**.

📊 Visual Diagram: Asymmetric Financial Risk



💻 Python Implementation

```
def calculate_loan_portfolio_loss(n_fp, n_fn, principal_loss=100000, interest_loss
    """
```

```
    Demonstrates Asymmetric Financial Loss in Loan Underwriting:
    - False Positive (Type I): Bank approves a defaulter -> Loses entire principal
    - False Negative (Type II): Bank rejects a good borrower -> Loses opportunity

    False Positives are 10x more costly, explaining why banks prioritize Precision
    """
    total_fp_loss = n_fp * principal_loss
    total_fn_loss = n_fn * interest_loss
    total_financial_loss = total_fp_loss + total_fn_loss

    results = {
        'total_loss': total_financial_loss,
        'fp_loss_ratio': total_fp_loss / (total_financial_loss + 1e-9)
    }
    return results

# --- Function Call & Testing ---
# Suppose 10 defaulters were approved (FP) and 20 good borrowers rejected (FN)
loss_report = calculate_loan_portfolio_loss(n_fp=10, n_fn=20)
print(f"Total Portfolio Loss: Rs {loss_report['total_loss']:,}")
print(f"Loss Driven by False Positives: {loss_report['fp_loss_ratio']*100:.1f}%")
```

🔑 Key Takeaways

1. In retail lending, False Positives are catastrophic Non-Performing Assets (NPAs).
2. Because of this asymmetry, the CreditWise system optimizes primarily for **Precision**.